

Development: CrowdStance — Cross-Platform Opinion Retrieval and Stance Analysis

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ABSTRACT

CrowdStance is an information retrieval system that aggregates and analyzes community opinions across Reddit and Hacker News. Given a search query, the system retrieves top comments from both platforms, classifies each comment as For, Against, or Nuanced using a stance detection model, clusters similar arguments using semantic embeddings, ranks them by engagement and confidence, and surfaces cross-platform divergence. The result is a structured, stance-aware view of how different online communities perceive the same topic. CrowdStance addresses the challenge of information overload in online discourse by transforming unstructured comment data into interpretable opinion summaries. The system is built with a Python and Flask backend, a React frontend, and integrates the Reddit and Hacker News APIs for live data retrieval.

CCS CONCEPTS

- Information systems → Information retrieval; Sentiment analysis; Query representation

KEYWORDS

Information Retrieval, Stance Detection, Opinion Aggregation, Semantic Embeddings, Reddit, Hacker News, Cross-Platform Analysis

1. INTRODUCTION

Online communities such as Reddit and Hacker News generate millions of comments daily on topics ranging from technology and politics to health and culture. While these platforms offer rich, diverse perspectives, extracting meaningful insight from the noise is difficult. A user searching for opinions on a topic must manually browse threads, synthesize arguments, and judge credibility, a time-consuming and cognitively demanding process.

CrowdStance addresses this problem by automating opinion retrieval and analysis across platforms. The system takes a natural language query, retrieves relevant comments from Reddit and Hacker News, classifies each comment by stance, clusters similar arguments, and presents the results in a structured, readable interface. Users can immediately see how a topic is perceived across communities, which arguments are most common, and where the two platforms diverge.

The intended users are researchers, journalists, students, and curious individuals who want a quick, structured view of public opinion on any topic. CrowdStance is particularly useful for understanding contested topics where community context matters.

2. SYSTEM OVERVIEW

CrowdStance is built around a five-stage pipeline: query processing, data retrieval, stance detection, argument clustering, and result presentation.

When a user submits a query, the system first expands it into aspects and subtopics using a language model, giving users the option to refine their search before retrieval begins. The retrieval stage fetches the top posts and comments from Reddit via the PRAW library and from Hacker News via the Algolia search API. Comments are filtered for minimum length and relevance before being passed to the analysis pipeline.

Each comment is then classified into one of three stance categories: For, Against, or Nuanced — using zero-shot classification with a pre-trained natural language inference model. Comments are also assigned a confidence score reflecting the strength of the classification. Semantically similar comments are grouped into clusters using sentence embeddings, and each cluster is ranked by a composite score combining engagement signals and confidence.

Finally, the system computes a cross-platform divergence score by comparing the stance distributions of Reddit and Hacker News, highlighting topics where the two communities disagree most strongly.

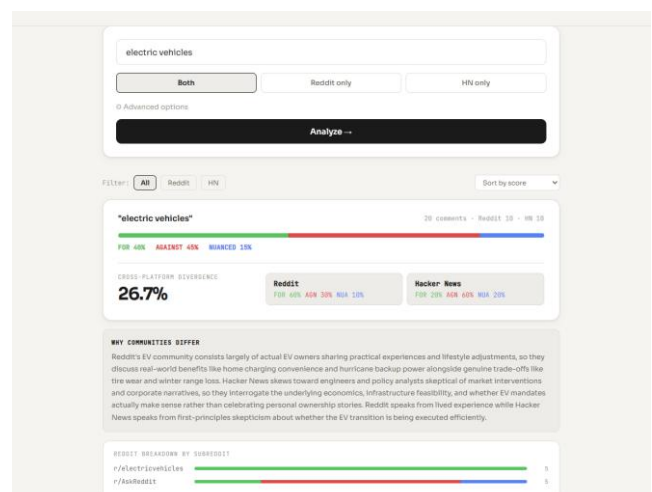


Figure 1: Stance distribution and cross-platform divergence

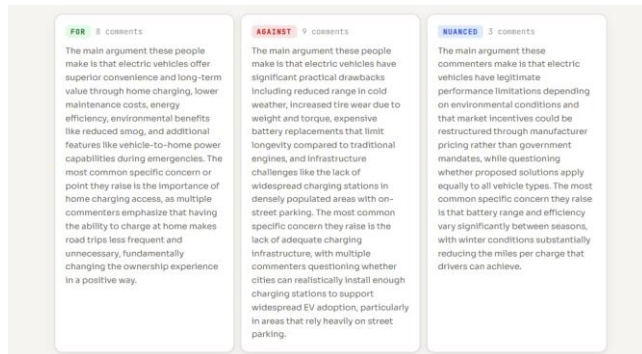


Figure 2: Insight cards per stance category

3. IMPLEMENTATION

CrowdStance is implemented as a full-stack web application. The backend is built with Python and Flask, and the frontend is built with React. The two communicate via a REST API.

3.1 Data Retrieval

Reddit data is retrieved using the PRAW library, which provides access to post titles, comment threads, upvote counts, and subreddit metadata. Hacker News data is retrieved using the Algolia HN Search API, which supports full-text search over stories and comments. For each query, the system retrieves the top 50 comments per platform, filtered to a minimum of 20 characters to remove low-quality responses.

3.2 Stance Detection

Stance classification is performed using the facebook/bart-large-mnli model [1] via the Hugging Face transformers library. Each comment is passed through a zero-shot classification pipeline with the candidate labels For, Against, and Nuanced. The model returns a probability score for each label, and the highest scoring label is assigned as the stance. The confidence score is the probability of the winning label.

3.3 Argument Clustering

Semantic embeddings are generated for each comment using the all-MiniLM-L6-v2 [2] sentence transformer model. Comments are clustered using agglomerative clustering with a cosine distance threshold, grouping semantically similar arguments together. Each cluster is then ranked by a composite score combining the average upvote count and average confidence of its members.

3.4 Cross-Platform Divergence

The divergence score is computed by comparing the stance distributions of Reddit and Hacker News using Jensen-Shannon divergence. A high score indicates that the two communities hold meaningfully different views on the topic.

3.5 Frontend

The React frontend provides an interactive results page with a stance bar showing the proportion of For, Against, and Nuanced comments, insight cards summarizing the top arguments per stance, a subreddit breakdown panel, confidence distribution chart, and a compare mode for analyzing two topics side by side.

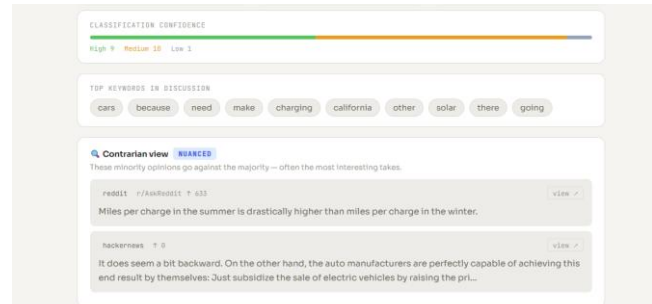


Figure 3: Classification confidence and contrarian view

4. HOW TO USE

To use CrowdStance, navigate to the application in a web browser. On the landing page, enter a topic or question into the search bar and click Analyze. The system will display a query intelligence panel showing detected aspects of the topic. Users can select specific aspects to narrow the analysis or proceed with the full query.



Figure 4: CrowdStance landing page

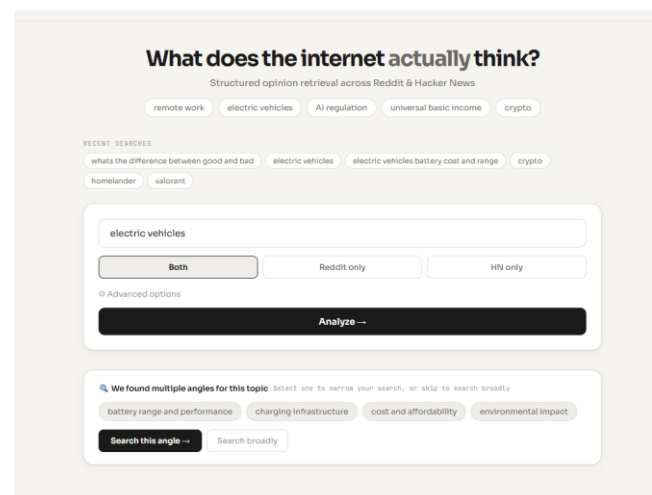


Figure 5: Query intelligence panel with detected aspects

The results page displays a stance bar at the top showing the overall distribution of For, Against, and Nuanced comments across both platforms. Below this, insight cards present the strongest arguments for each stance category. The subreddit breakdown panel shows which communities are most active on the topic and their individual stance distributions.

Users can expand individual comment cards to view the full comment text and the reasoning behind its stance classification. The confidence distribution chart shows how certain the model was across all classifications. For advanced analysis, users can enable Compare Mode by entering a second topic, which displays both topics side by side for direct comparison. The divergence score at the top of the results indicates how differently Reddit and Hacker News communities perceive the topic.



Figure 6: Comment card showing stance reasoning

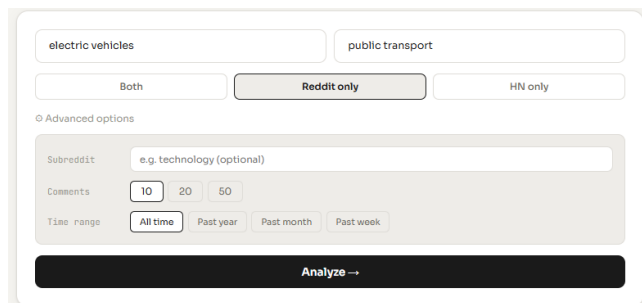


Figure 7: Advanced options panel

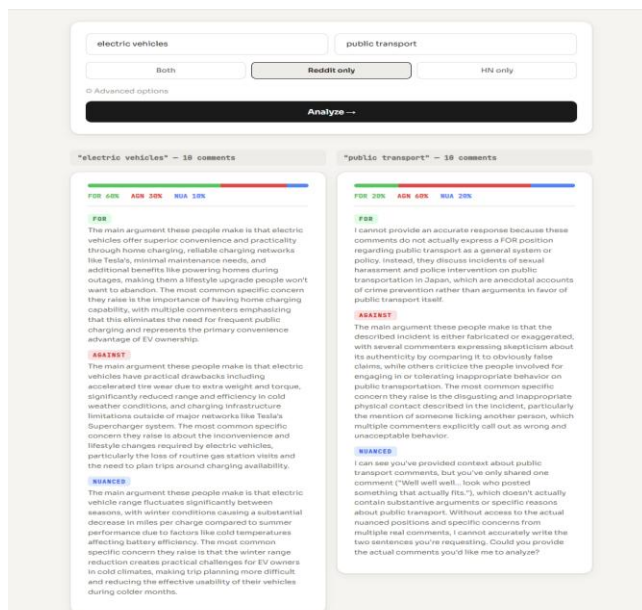


Figure 8: Compare mode showing two topics side by side

5. EVALUATION

CrowdStance was evaluated through functional testing and qualitative analysis of results across multiple queries.

5.1 Functional Testing

A test suite of 33 test cases was developed to verify the correctness of all major system components. Tests covered API connectivity, comment retrieval, stance classification, clustering, divergence computation, and frontend rendering. All 33 tests pass successfully, confirming that the system functions correctly end to end.

```
=== CrowdStance Test Suite ===

✓ PASS — Health endpoint
✓ PASS — Health returns status ok
✓ PASS — Frontend loads
✓ PASS — Frontend is HTML
✓ PASS — Empty query rejected

Running analysis for 'remote work' (may take 20-30s)...
✓ PASS — Analyze endpoint returns 200
✓ PASS — Response has query field
✓ PASS — Response has grouped field
✓ PASS — Response has FOR/AGAINST/NUANCED
✓ PASS — Response has divergence
✓ PASS — Response has summaries
✓ PASS — Response has confidence_distribution
✓ PASS — Response has top_keywords
✓ PASS — Total comments > 0
✓ PASS — No error in response

Testing cache (same query again)...
✓ PASS — Cache hit works
✓ PASS — Intelligence endpoint returns 200
✓ PASS — Intelligence returns core_topic
✓ PASS — Intelligence returns aspects
✓ PASS — Empty intelligence query rejected

Testing Reddit-only search...
✓ PASS — Reddit-only returns results
✓ PASS — Reddit-only has no HN comments

Testing HN-only search...
✓ PASS — HN-only returns results
✓ PASS — HN-only has no Reddit comments

Testing compare endpoint...
✓ PASS — Compare endpoint returns 200
✓ PASS — Compare returns a and b
✓ PASS — Compare a has query
✓ PASS — Compare b has query
✓ PASS — Unknown topic handles gracefully

Testing subreddit filter...
✓ PASS — Subreddit filter works

=== Tests Complete ===
```

Figure 9: CrowdStance test suite — 33/33 tests passing

5.2 Qualitative Analysis

The system was tested on a range of topics including electric vehicles, remote work, artificial intelligence, and social media regulation. For each topic, CrowdStance successfully retrieved comments from both platforms, classified stances, and generated insight summaries. Results were manually reviewed and found to be coherent and representative of actual community sentiment.

Cross-platform divergence scores varied meaningfully across topics. Technical topics such as programming languages showed higher divergence between Reddit and Hacker News, reflecting the different demographics and expertise levels of each platform. General topics such as remote work showed more agreement across platforms.

5.3 Limitations

The system is limited by the quality and availability of API data. Reddit's API rate limits the number of comments retrievable per query. Stance classification accuracy depends on the zero-shot model and may degrade on highly sarcastic or context-dependent comments. Future work includes fine-tuning the stance model on Reddit-specific data and expanding to additional platforms.

6. CONCLUSION

CrowdStance demonstrates that information retrieval techniques can be effectively applied to the problem of online opinion aggregation. By combining live data retrieval, zero-shot stance detection, semantic clustering, and cross-platform divergence analysis, the system transforms raw comment data into structured, interpretable opinion summaries.

The system successfully retrieves and analyzes community opinions in real time, presenting results in an interface that is both informative and accessible. CrowdStance fills a genuine gap, existing platforms either show raw comments with no structure, or aggregate sentiment without preserving argument diversity. CrowdStance does both.

Future development will focus on expanding platform coverage to include Twitter and YouTube, improving stance detection accuracy through fine-tuning on domain-specific data, and adding user accounts to allow saved searches and longitudinal tracking of opinion shifts over time.

The source code is available at:

<https://github.com/lokstyle/crowdstance>

7. CONTRIBUTIONS

Lokesh Ravi designed and developed the complete CrowdStance system, including the backend API, stance detection pipeline, argument clustering module, cross-platform divergence computation, React frontend and test suite.

ACKNOWLEDGMENTS

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develop this project. Reddit and Hacker News APIs were used in accordance with their respective terms of service.

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